Permutations with Indistinguishable Objects

- How many different strings can be made by reordering the letters of the word *SUCCESS*?
- **Solution:**
  Since some of the letters repeat in SUCCESS, so the number of permutations for 7 distinct letters is not the correct answer here. SUCCESS contains 3 S's, 2 C's, 1 E and 1 U. The three indistinguishable S's can be placed among the 7 positions in $C(7,3)$ different ways, leaving 4 positions free. Then, the two C's can be placed in $C(4,2)$ ways, leaving 2 free positions. … By the product rule, the number of different ways is

$$C(7,3)C(4,2)C(2,1)C(1,1) = \frac{7!}{3!\,2!\,1!\,1!}$$